

Checkpoint 4 — Newton–Raphson Update for Bayesian Logistic Regression — Student Version

INFO 521 — Machine Learning Foundations · checkpoint (prompts & criteria; graded in class)

i How this works

This checkpoint is sat **in class, closed-book** (~15–20 minutes, one derivation) and graded Satisfactory / Not-Yet against the criteria below. Version B is the alternate-version retake, offered the following week; you keep your best result. Knowing the prompt in advance is the point — practice until you can reproduce the derivation unaided.

Sitting: Week 5 · **retake:** Week 6 · **gates** Project 2 · **Gate A.**

Gates: Project 2.1 (Approximate Inference) — and with it, Gate A. **Format:** in-class, closed-book, ~15–20 min, one derivation. Graded Satisfactory / Not-Yet. **Retakeable** (an alternate version is provided below).

This checkpoint tests the study-guide outcome “*derive a second-order optimizer for a non-conjugate posterior.*” It is the supervised-gate counterpart to Checkpoints 1–2: where those derived the closed-form least-squares/MLE solution, this one derives the **iterative** update you need once conjugacy is gone.

Prompt (Version A)

Bayesian logistic regression places a Gaussian prior $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{S}_0)$ on the weights, with a Bernoulli likelihood $y_n \sim \text{Bernoulli}(\mu_n)$, $\mu_n = \sigma(\mathbf{w}^\top \mathbf{x}_n)$ and σ the logistic sigmoid. Writing $E(\mathbf{w}) = -\log p(\mathbf{w} | \mathcal{D})$ (up to a constant), derive (i) the **gradient** ∇E , (ii) the **Hessian** $\nabla^2 E$, and (iii) the **Newton–Raphson update** for the MAP estimate. State why the update is called *iteratively reweighted least squares*.

Prompt (Version B — retake)

Explain the **Laplace approximation** of the posterior in this same model: what it is, *why* the **negative Hessian of the log-posterior at the mode** is the posterior **precision**, and give **one** situation in which the approximation is poor.

Satisfactory criteria

- (A) Correct ∇E and $\nabla^2 E$, including the prior term $\mathbf{S}_0^{-1} \mathbf{w}$ (gradient) and $\mathbf{S}_0^{-1}$ (Hessian).
- (A) States the update $\mathbf{w} \leftarrow \mathbf{w} - \mathbf{H}^{-1} \mathbf{g}$ and identifies the reweighting matrix $\mathbf{R} = \text{diag}(\mu_n(1 - \mu_n))$ as the “weights” that make it reweighted least squares.
- (B) Names the Gaussian centred at the mode; argues the precision = curvature of the log-posterior at the mode; gives one valid failure case (skewed / multimodal / near-separable posterior).